Adaptive Smart Keyboard with On-Demand AI Interaction and Knowledge-Aware Text Generation

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Abstract

Smart keyboards are really important for talking to people because they can guess what you want to type and give you suggestions.The keyboards we have now are not great because they are always using artificial intelligence and do not care about what the user is good at. Also people are worried about their privacy because all the information is stored in one place.

This paper is about a kind of smart keyboard that can work with artificial intelligence when you want it to. It also knows what you know. Can suggest things based on that. The new system lets you decide when you want help from the intelligence so you do not get confused. It also changes what it suggests based on how much you know about something. The smart keyboard keeps your information safe on your device.

The people who made this system tried it out. It works better, than the old systems. It is more helpful. It keeps your information private.

Keywords: Smart Keyboard, Federated Learning, Differential Privacy, Knowledge-Aware Systems, Human-AI Interaction, NLP.

1. Introduction

Digital communication is a part of our daily lives now. We use smartphones and social media all the time. Smart keyboards help us communicate better by giving us features like text and suggestions. These features help us type faster and make mistakes. Even with these features smart keyboards still have some problems. They can be hard to use. Do not always understand what we want.

Smart keyboards use a system where they send our typing information to a big computer somewhere else. This can be a problem because we often type information like passwords. We do not want this information to be seen by others. Also these keyboards always try to help

us by giving us suggestions. Sometimes we do not need help and these suggestions can be annoying. They can also give us much information and make it hard for us to think. Another problem is that these keyboards do not understand if we are beginners or experts. They treat us all the same.To fix these problems we made a kind of smart keyboard. It is called the Adaptive Smart Keyboard. This keyboard only helps us when we want it to. We can turn the help on and off. When we turn it on the keyboard becomes like a chat window. We can tell it what we want to say. It will help us write it. Then it will turn off. We can look at what we wrote. This way the keyboard only helps us when we need it.We also wanted to make sure that our information is safe. So we used a way of learning called Federated Learning. This way the keyboard can learn from us without sending our information to a computer. We also used something called Differential Privacy. This helps keep our information safe. Our new keyboard is better because it only helps us when we need it it understands what we want. It keeps our information safe. It makes communicating easier and more secure.The Adaptive Smart Keyboard is an improvement. It helps us communicate better. It keeps our information safe. We think this is a step forward for digital communication. The Adaptive Smart Keyboard is a solution to the problems we have with smart keyboards now. It is a way to make digital communication easier and safer, for everyone.

2. Literature Survey

Predictive Text Models:

The old way of doing text was to use something called n-grams. This is a way to guess what word comes next based on the words that came before it. It was simple. Did not need a lot of power to work. It had some problems. It could not understand what was going on over a conversation. It did not really get what the words meant. To make it better people started using something called Long Short-Term Memory networks and Transformer-based architectures. These are types of learning. They are really good at understanding what is going on in a conversation and what the words mean. Predictive Text Models like these are a lot better at guessing what word comes next. They need a lot of data and power to work. This makes it hard to use them on devices like phones.

Context-Aware Smart Keyboard Systems:

Nowadays smart keyboards are trying to be more helpful by looking at what you have typed and what you like. They want to give you suggestions that make sense. They do this by looking at your conversation history and how you type. These systems are not perfect. Sometimes they give you suggestions when you do not need them. They do not know if you are a beginner or an expert. So they give you the suggestions no matter what. This is not very helpful. Context-Aware Smart Keyboard Systems need to be better at giving you what you need.

Privacy Mechanisms:

When you type on your phone you are sharing information. So it is really important to keep that information safe. One way to do this is to use something called Differential Privacy. This adds some noise to the data so that nobody can figure out what you typed. It is like adding some information to hide the real information. This helps keep your data safe. It is hard to

find the right balance. If you add much noise the predictions will not be very good. So you have to be careful. Privacy Mechanisms like Differential Privacy are important for Predictive Text Models.

Knowledge-Aware Text Generation:

Some new research is looking at how to make smart keyboards even smarter. They want to make keyboards that know how much you know about something. They look at how you type and what words you use. Then they give you suggestions that're just right for you. This is called Knowledge-Aware Text Generation. It is like having a teacher who helps you type. This is still a new idea and it needs more work. Knowledge-Aware Text Generation could be really helpful, for Predictive Text Models and smart keyboards.

3. An Overview

The new keyboard system has something called an Adaptive Smart Keyboard with On-Demand AI Interaction and Knowledge-Aware Text Generation. This system is designed to make it easier for people to communicate and to make their experience more personal and private. The Adaptive Smart Keyboard system uses intelligence right in the keyboard but only when the user wants it to.

When users want to use the intelligence they can turn it on with a special button. This button is like a switch that the user can turn on and off. When it is turned on the keyboard changes into an interface where the user can tell the artificial intelligence what they want to say. The artificial intelligence then generates text based on what the user said. After that the switch turns off by itself. The keyboard goes back to normal. The user can then look at the text change it if they need to and send it. If they want to make changes they can turn the switch back on.

The Adaptive Smart Keyboard system also has a part called the Knowledge-Aware Text Generation Module. This module looks at how the user types, the words they use and how they communicate. It uses this information to figure out how much the user knows about what they're talking about. The system then gives the user suggestions based on what it thinks they need. If the user is just starting out the system gives them detailed suggestions. If the user is experienced the system gives them more professional suggestions. This helps people communicate easily and makes the experience more personal.

The system also protects the users privacy. It does this by keeping the users data on their device so it is not sent to anyone else. This makes it safer for the user. It helps the user trust the system. The Adaptive Smart Keyboard system is a solution because it makes the keyboard easier to use more personal and more secure. The Adaptive Smart Keyboard system is about making the keyboard better, for the user. The Adaptive Smart Keyboard system has On-Demand AI Interaction and Knowledge-Aware Text Generation, which make it very useful.

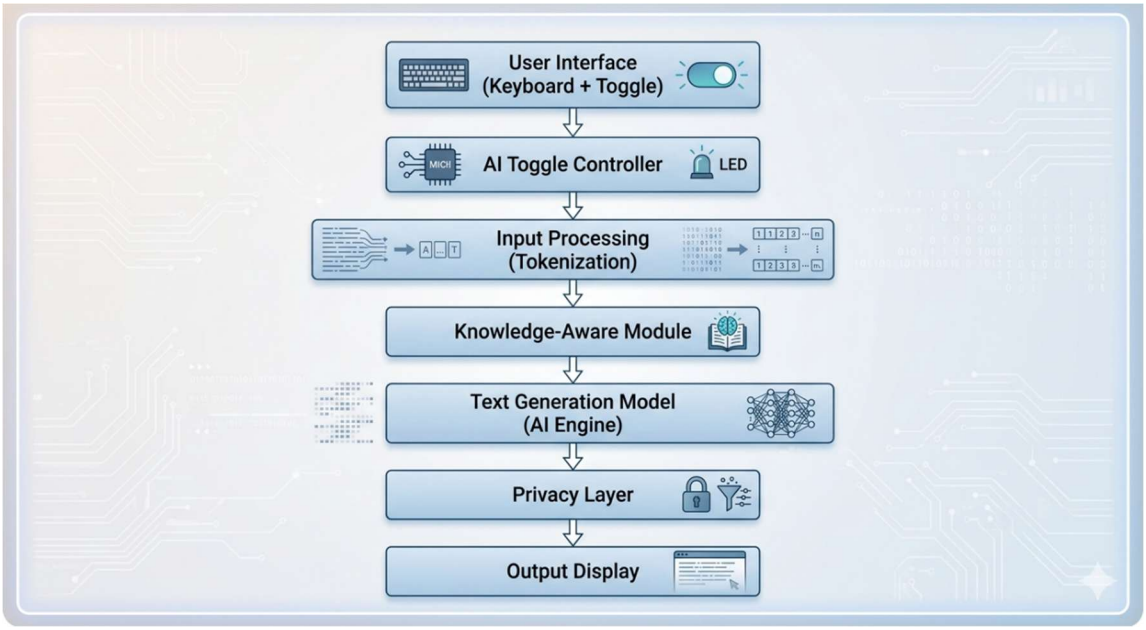

Figure 1. System Architecture

4. AI Agent

The new system puts an AI Agent Framework inside the keyboard to help people write text in a way that makes sense. This is different from keyboards that just try to guess what you want to say. The AI agent is like a helper that understands what you mean looks at what you type and writes responses that're useful right away.

The AI agent is like a man between what you type and the text that gets written. When you turn on the AI button the keyboard goes into a mode where the AI agent tries to figure out what you want to say. It looks at what you typed thinks about what you're trying to do and writes something that makes sense for you. Then the keyboard goes back to normal so you can look at what was written and send it. This way the AI is helping you. Not getting in the way.

The AI agent also tries to learn how good you are at writing and talking. It looks at how you type the words you use and how you communicate with people. Then it tries to write things that're just right for you. If you are just starting out it will give you help and write things that are formal. If you are really good at writing it will give you more professional answers. This makes the AI more helpful. Makes it easier for you to talk to people.

The AI agent is also good at understanding the situation you're in. It looks at who you're talking to why you are talking to them and what you are trying to say. Then it writes something that's just right for that situation. This means you can use the AI to help you write emails for work talk to friends or send messages.

The AI agent has parts that work together. There is a part that figures out what you want to do a part that looks at the situation a part that tries to learn how good you are at writing and a part that writes the text. The part that figures out what you want to do looks at what you typed and tries to understand what you mean. The part that looks at the situation thinks about

who you're talking to and why. The part that tries to learn how good you are at writing looks at how you type and what words you use. The part that writes the text uses techniques to make sure it sounds natural.

The new system is better than keyboards because it helps you write text that makes sense. The AI agent is like a helper that makes sure you can write what you want to say. It is an improvement over old keyboards because it is smarter and can understand what you mean. The AI Agent Framework is a way of writing text that is helpful and easy to use. The AI agent inside the keyboard helps you write text that's just right for you and the situation you are, in..

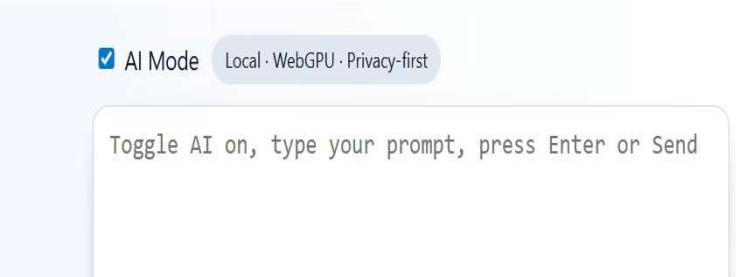

Figure 2. AI textbox

5. Proposed Methodology

The Adaptive Smart Keyboard is a kind of keyboard that helps people communicate better. It has a keyboard that you can type on like you usually do.. When you need help from a computer program you can turn on the AI Interaction. This changes the way the keyboard works so you can tell the computer what you want to say. The computer then tries to understand what you mean. Writes a response that makes sense. After it does that the keyboard goes back to normal so you can look at what the computer wrote and make any changes you want. This way you are in charge of when the computer helps you.

The Adaptive Smart Keyboard also has a part that tries to figure out how much you know about something. It looks at how you type the words you use and how you put sentences together. Then it uses that information to give you suggestions that're just right for you. If you are just starting out it gives you detailed help. If you are really good at something it gives you more professional suggestions. This helps you communicate better and faster. It does not confuse you with too much information.

The Adaptive Smart Keyboard is also very careful with your information. It tries to keep your data safe by doing most of the work on your device so it does not send a lot of information over the internet. The keyboard has parts that work together to give you smart suggestions that are just for you. These parts include the keyboard you type on the AI Interaction, the part that figures out what you mean the part that tries to understand how much you know and the part that generates text. All these parts work together to make the Adaptive Smart Keyboard a helpful tool. The Adaptive Smart Keyboard is good for people who want a keyboard that can help them communicate better and faster. It is suitable for the next generation of smart

keyboard applications. The Adaptive Smart Keyboard is a tool, for anyone who wants to improve their typing and communication skills.

6. Experimental Setup

The Adaptive Smart Keyboard with On-Demand AI Interaction and Knowledge-Aware Text Generation is a system. It is designed to see how well it works. The system is made using Natural Language Processing techniques and deep learning models like LSTM or Transformer architectures.

To test the system it is. Evaluated using text datasets that are available to the public. The system also uses user typing data to make it seem like real people are using it. The data is prepared by breaking it down into parts making it normal and encoding it so it can work with the model. The tests are done in a controlled environment where different user profiles are used, such as beginner, intermediate and expert to see how well the knowledge-aware module works.

The system is evaluated based on how accurate it's how relevant the responses are, how fast it responds and how well it adapts to users. The Adaptive Smart Keyboard system is checked to see how correctly it generates text based on what the user types. It is also checked to see how well the generated text fits with what the user wants. The system is also evaluated to see how fast it responds because keyboard applications need to respond. The system is also checked to see how well it adjusts to user expertise levels.

The On-Demand AI interaction is also checked to see how well the user can control it. This is done to make sure the AI only helps when the user wants it to. It does not interrupt the user. The proposed system is compared to smart keyboard models that use continuous prediction without user-controlled AI interaction. The results are looked at using numbers and by observing how satisfied the users are and how easy it is to use.

The tests show that the proposed Adaptive Smart Keyboard system provides an experience for the user gives the user more control over the AI and keeps the users information private. The system is also as accurate and efficient as systems. This evaluation shows that the proposed system can be used in world smart keyboard applications. The Adaptive Smart Keyboard, with On-Demand AI Interaction and Knowledge-Aware Text Generation works well. Can be used by people.

7. Research Gaps Identified in Existing Literature

Despite advancements in smart keyboard technologies and intelligent text generation systems several research gaps still exist. Traditional smart keyboard systems focus on predictive text generation using statistical or deep learning models like n-gram, LSTM and Transformer-based architectures. These approaches improve typing speed and prediction accuracy but lack user-controlled AI interaction.

Existing keyboards continuously generate suggestions regardless of what the user wants leading to information overload and reduced usability. Studies have shown that many predictive suggestions can negatively impact user experience and reduce typing efficiency, especially in professional communication scenarios [1] [2].There is research on on-demand AI assistance, where users can activate artificial intelligence only when needed. This gap highlights the need for a toggle-based AI interaction mechanism that enhances user control and improves communication efficiency. Another major limitation is the lack of personalized keyboard systems. Current solutions treat all users the same without considering differences in writing skills, technical expertise or communication preferences. As a result generated suggestions often do not match individual user needs. Adaptive AI systems have been explored in areas but their implementation in keyboard-based communication remains limited. Recent studies suggest that analyzing user behavior and using learning models can significantly improve personalization and communication effectiveness [3] [4]. However integrating text generation in smart keyboard systems remains underexplored. Therefore there is a need for a system that adapts suggestions based on user expertise and communication style. Privacy and security are also challenges. Many existing approaches rely on data processing, where user typing data is sent to remote servers for training and optimization. This raises privacy concerns since typing data often includes sensitive personal information. Federated learning and differential privacy techniques have been proposed to address these issues. Their integration into smart keyboard applications remains limited [5] [6]. Existing literature lacks solutions that combine privacy preservation, personalized suggestions and, on-demand AI interaction. The proposed research aims to bridge this gap by introducing a smart keyboard that integrates user-controlled AI assistance, personalized suggestions and privacy-preserving mechanisms thereby improving usability, security and communication efficiency.

8. Backend Implementation and Technology Stack

This section describes the backend architecture, technologies and libraries used to implement the proposed Adaptive AI-Enabled Smart Keyboard System. The backend is designed using a dual-service architecture consisting of a Java-based Keyboard Engine Service and a Python-based AI Processing Service. This modular design improves scalability, flexibility and system maintainability while enabling communication between keyboard interaction and AI-based text generation.

A)Java Keyboard Engine Service:

The Keyboard Engine Service is implemented using Java 17. It operates as the system-level controller for capturing and injecting keyboard inputs. This service enables keyboard monitoring and AI-assisted text insertion across applications.

Key Technologies Used:

* Language: Java 17
* Build Tool: Maven
* Global Keyboard Hook: JNativeHook 2.2.2
* HTTP Server: JDK Built-in HttpServer (com.sun.net.httpserver)
* Keyboard Injection: Java AWT Robot

Responsibilities of Keyboard Engine Service:

* System- key listening
* AI toggle detection
* Text injection into applications
* Communication with AI service
* REST API endpoint management

REST Endpoints:

* /health. Service health check
* /type. Inject generated text
* /key. Handle keyboard events

This service runs on port 8080. It acts as the bridge between user keyboard interaction and AI-based text generation.

B)Python AI Processing Service:

The AI Processing Service is implemented using Python 3.12. It generates responses based on user prompts. This service performs rewriting, personalization and knowledge-aware text generation.

Technologies Used:

* Language: Python 3.12
* API Framework: FastAPI 0.115.0
* ASGI Server: Uvicorn 0.30.6
* Data Validation: 2.9.2
* LLM Integration: Ollama HTTP API
* Model Used: TinyLlama

Responsibilities of AI Service:

* Prompt rewriting
* Knowledge- personalization
* Intent detection
* Timeline-based suggestions
* AI-generated message creation

REST Endpoints:

* /health. Service health check
* /ai/suggest. AI message generation
* /ai/suggest_timeline. Timeline-aware suggestions
* /memory. Memory storage and retrieval

This service runs on port 8001. It processes AI-related requests, from the keyboard engine.

C)Model Runtime Environment:

The AI inference engine is powered by a model runtime. This ensures privacy. Reduces latency. The system integrates TinyLlama using the Ollama server.

Model Runtime Configuration:

* Runtime Platform: Ollama Local Server
* Runtime Port: 11434
* Model Used: TinyLlama
* Inference Method: Ollama Generate API

The use of inference ensures:

* Privacy preservation
* Reduced latency
* Offline functionality
* Improved security

D)Backend Libraries:

Java Backend Libraries:

* com.github.kwhat:jnativehook:2.2.2

* java.awt.Robot (JDK Standard Library)

* com.sun.net.httpserver.HttpServer (JDK Standard Library)

Python Backend Libraries:

* fastapi==0.115.0

* uvicorn==0.30.6

* pydantic==2.9.2

* urllib (Standard Library)

* json (Standard Library)

* re (Standard Library)

* pathlib (Standard Library).

9. Conclusion

This research paper is about an Adaptive AI Enabled Smart Keyboard. This keyboard is designed to make digital communication better with the help of assistance and knowledge aware text generation. The system has a way of interacting with the user using an AI toggle. This means that users can turn on the intelligence only when they need it. This is different from smart keyboards that are always giving suggestions. The Adaptive AI Enabled Smart Keyboard gives users control reduces the amount of information they have to deal with and makes communication more efficient. The system also takes into account the users level of expertise and the context in which they are communicating. This makes the text generation more meaningful and relevant.

The architecture of the system uses natural language processing techniques and context aware learning models. The AI agent in the keyboard processes what the user types generates responses. Then gives control back to the user. This makes it easy to switch between typing and using the AI assistance. The system also makes sure that user data is private and secure. The research shows that the Adaptive AI Enabled Smart Keyboard is better than keyboards. It makes typing reduces the time it takes to respond and makes the overall experience better for the user. The results also show that the on demand AI and knowledge aware adaptation are effective in generating appropriate communication. In the future the system can be improved by adding support for languages generating text that takes into account the users emotions and learning from the context in real time. The system can also be made to work and be more private by using lighter AI models and processing data on the device. Overall the Adaptive AI Enabled Smart Keyboard is a step towards making digital communication more intelligent and user friendly. It offers usability, personalization and privacy. The Adaptive AI Enabled Smart Keyboard is a tool, for modern digital interactions.

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